

Multiple Choice (10 marks)

1. The quantum efficiency of a photon detector is 0.1. If 100 photons are sent into the detector, one after the other, the detector will detect photons
 - (a) exactly 10 times
 - (b) an average of 10 times, with an rms deviation of about 0.1
 - (c) an average of 10 times, with an rms deviation of about 1
 - (d) an average of 10 times, with an rms deviation of about 3
2. Which of the following ions CANNOT be used as a dopant in germanium to make an n-type semiconductor?
 - (a) As
 - (b) P
 - (c) Sb
 - (d) B
3. A satellite of mass m orbits a planet of mass M in a circular orbit of radius R . The time required for one revolution is
 - (a) independent of M
 - (b) proportional to $m^{(1/2)}$
 - (c) linear in R
 - (d) proportional to $R^{(3/2)}$
4. A three-dimensional harmonic oscillator is in thermal equilibrium with a temperature reservoir at temperature T . The average total energy of the oscillator is
 - (a) $(1/2) k T$
 - (b) kT
 - (c) $(3/2) k T$
 - (d) $3 kT$
5. By definition, the electric displacement current through a surface S is proportional to the
 - (a) rate of change of the electric flux through S
 - (b) electric flux through S
 - (c) time integral of the magnetic flux through S
 - (d) rate of change of the magnetic flux through S

True / False (10 marks)

Answer True or False

6. True or False: If a photon detector has a quantum efficiency of 0.1, it will detect an average of 10 photons with an rms deviation of about 3.
7. True or False: Electromagnetic radiation from a nucleus is typically emitted as visible light.
8. True or False: An observer must move at $0.60\ c$ parallel to a $1.00\ \text{m}$ rod to measure its length as $0.80\ \text{m}$ due to length contraction.
9. True or False: The electric displacement current through a surface is proportional to the rate of change of the electric flux through that surface.
10. True or False: The frequency of the next higher harmonic for this pipe is $262\ \text{Hz}$, which is the second harmonic for open pipes.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the energy dynamics involved in ionizing a helium atom, explaining how the total energy required to remove both electrons relates to the energy needed to remove just one.
12. Discuss the scenario where a tube of water is moving at $1/2\ c$ relative to a lab frame when a light beam enters it in the same direction. Calculate the speed of light in the water relative to the lab frame, considering the index of refraction, and explain your reasoning.

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